



(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Academy of
Engineering

Department of E & TC Engineering

DIGITAL ELECTRONICS ENGINEERING LABORATORY [2307103L]

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TRAFFIC LIGHT CONTROLLER

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What is a Traffic Light Controller?

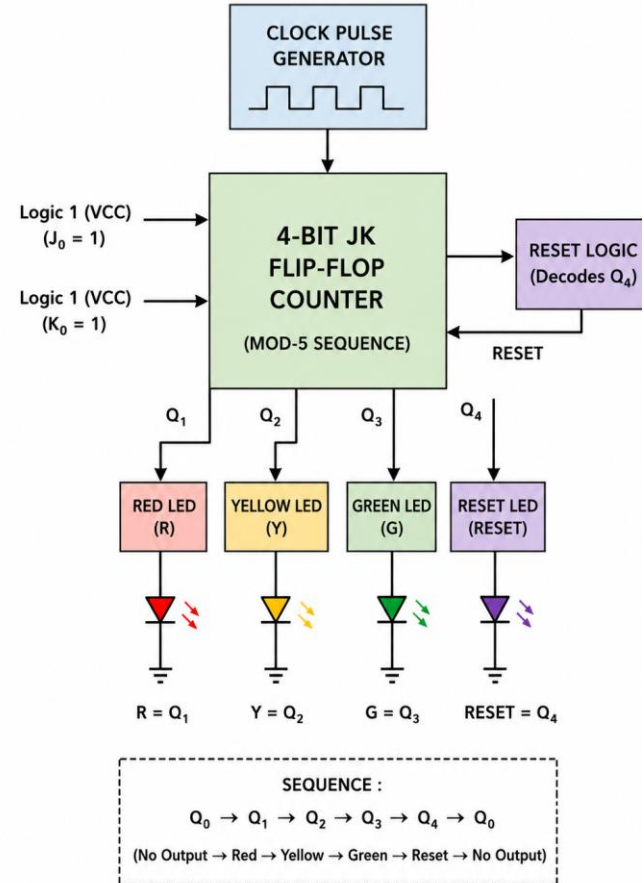
A Traffic Light Controller is an electronic system that automates signal sequencing at road intersections using digital logic circuits.

It controls Red, Yellow, and Green LEDs in a defined cyclic order to manage vehicle and pedestrian flow safely.

In this project, the CD4017 decade counter IC is used as the core controller, driven by a push-button clock pulse.

Importance & Applications

- Urban intersections — controls vehicle and pedestrian flow
- Railway level crossings — automated gate & signal control
- Smart city systems — IoT-integrated adaptive signaling
- Industrial zones — forklift and worker safety signaling
- Academic labs — demonstrates sequential digital logic



Problem Statement

Design and implement a low-cost, hardware-based Traffic Light Controller using the CD4017 decade counter IC that automatically sequences Red, Yellow, and Green LED signals in a cyclic order, simulating a real-world traffic signal system — without the need for a microcontroller or software programming.



Hardware Only

Implement complete signal sequencing using only digital ICs, resistors, LEDs, and passive components — no microcontroller.



Automatic Cycling

Ensure continuous automatic reset and re-cycling of the traffic light sequence from Red → Yellow → Green → Red.

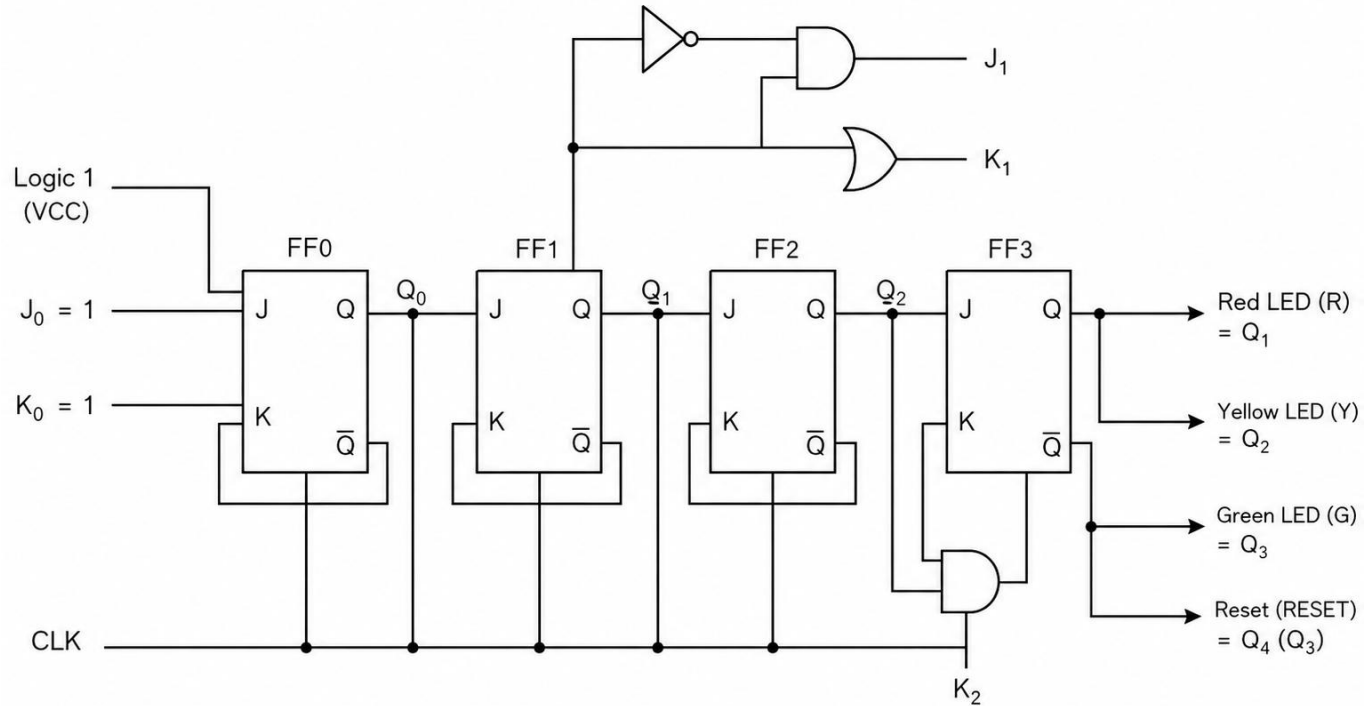


Low Cost & Simple

Use minimal, affordable, and easily available electronic components to keep the total project cost under ₹150.

Project Block Diagram

4



Sequence : $Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_3 \rightarrow Q_4 \rightarrow Q_0$
(No Output $\rightarrow$ Red $\rightarrow$ Yellow $\rightarrow$ Green $\rightarrow$ Reset $\rightarrow$ No Output)

Component Used

Component	Value / Model	Qty	Function
Decade Counter IC	CD4017 / 74HC4017	1	Core sequential counter — drives LED signals
LED (Red, Yellow, Green)	5mm Standard LED	3	Visual output for traffic signal states
Resistors	220Ω	3	Current limiting for each LED (Red, Yellow, Green)
Push Button	Manual	1	Clock pulse trigger for circuit control
Resistors	10kΩ	1	Pull-down resistor for reset pin stability
Battery	9v	1	Power supply for the entire circuit
Breadboard	Large	1	Prototyping platform for component connections
Diode	1N4007	3	Prevent back-current; isolate output signals

Truth Table

Q_0 — No output (Initial / Reset state)

Q_1 → Red LED ON (STOP)

Q_2 → Yellow LED ON (READY)

Q_3 → Green LED ON (GO)

Q_4 → RESET — counter returns to Q_0

K-Map Simplification

Signal	Min-term	Boolean Expression
Red (R)	$\Sigma m(1)$	$R = Q_1$
Yellow (Y)	$\Sigma m(3)$	$Y = Q_2$
Green (G)	$\Sigma m(7)$	$G = Q_3$
Reset	$\Sigma m(10)$	$RESET = Q_4$

Final Equations

$$R = Q_1 \quad Y = Q_2 \quad G = Q_3 \quad RESET = Q_4$$

→ Truth Table :-

CLK (Pulse No.)	Active Output	Q_0 (No Output)	Q_1 (Red)	Q_2 (Yellow)	Q_3 (Green)	Q_4 (Reset)
Initial (Power ON)	Q_0	1	0	0	0	0
1 st Pulse	Q_1	0	1	0	0	0
2 nd Pulse	Q_2	0	0	1	0	0
3 rd Pulse	Q_3	0	0	0	1	0
4 th Pulse	Q_4 (Reset)	0	0	0	0	1
5 th Pulse	Q_0	1	0	0	0	0

→ Logical Expressions (using K-map) :-

For Red LED ($R = Q_1$)					For Yellow LED ($Y = Q_2$)					For Green LED ($G = Q_3$)				
AB \ CD	00	01	11	10	AB \ CD	00	01	11	10	AB \ CD	00	01	11	10
00	0	1	0	0	00	0	0	1	0	00	0	0	0	0
01	0	0	0	0	01	0	0	0	0	01	0	0	1	0
11	0	0	0	0	11	0	0	0	0	11	0	0	0	0
10	0	0	0	0	10	0	0	0	0	10	0	0	0	0

$$R = \Sigma m(1) = Q_1$$

$$Y = \Sigma m(3) = Q_2$$

$$G = \Sigma m(7) = Q_3$$

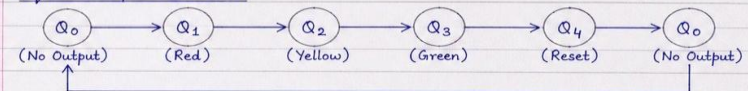
For Reset ($RESET = Q_4$)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	1

$$RESET = \Sigma m(10) = Q_4$$

Hence,
 $R = Q_1$
 $Y = Q_2$
 $G = Q_3$
 $RESET = Q_4$

→ Sequence Representation :-



Note :- After Q_4 (5th Output), counter resets and sequence starts again from Q_0 .

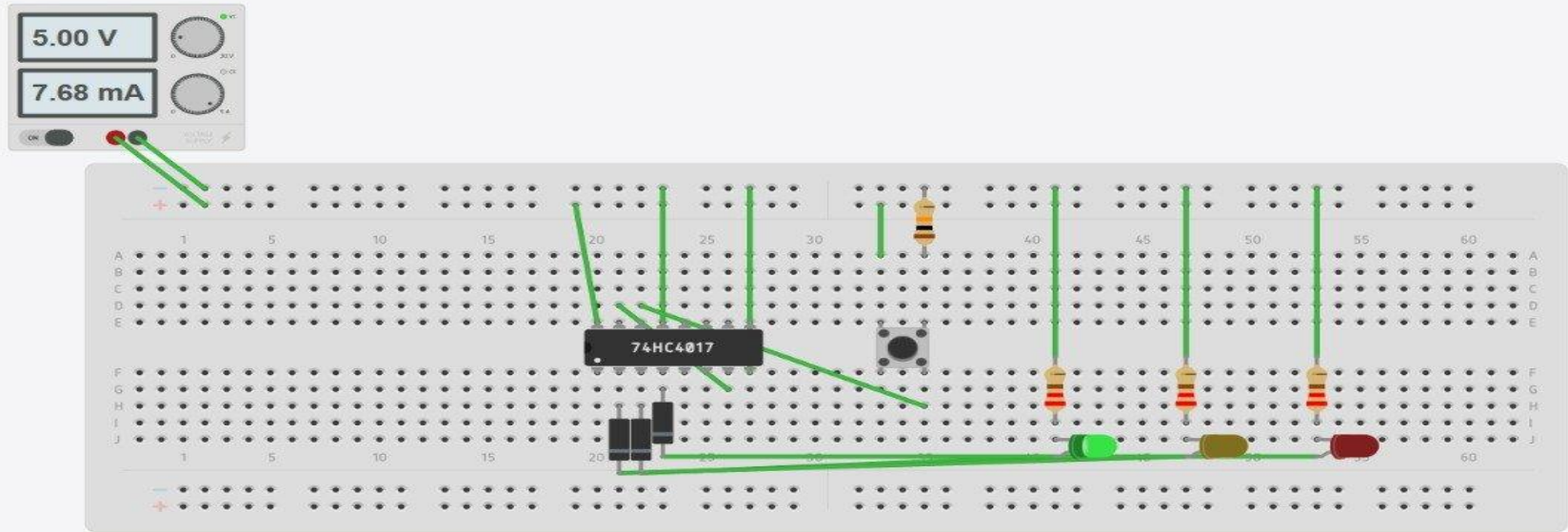


Fig: Tinkercad Breadboard Simulation — 74HC4017 Traffic Light Controller (5V, 7.68mA)

Supply Voltage

5.00 V

Supply Current

7.68 mA

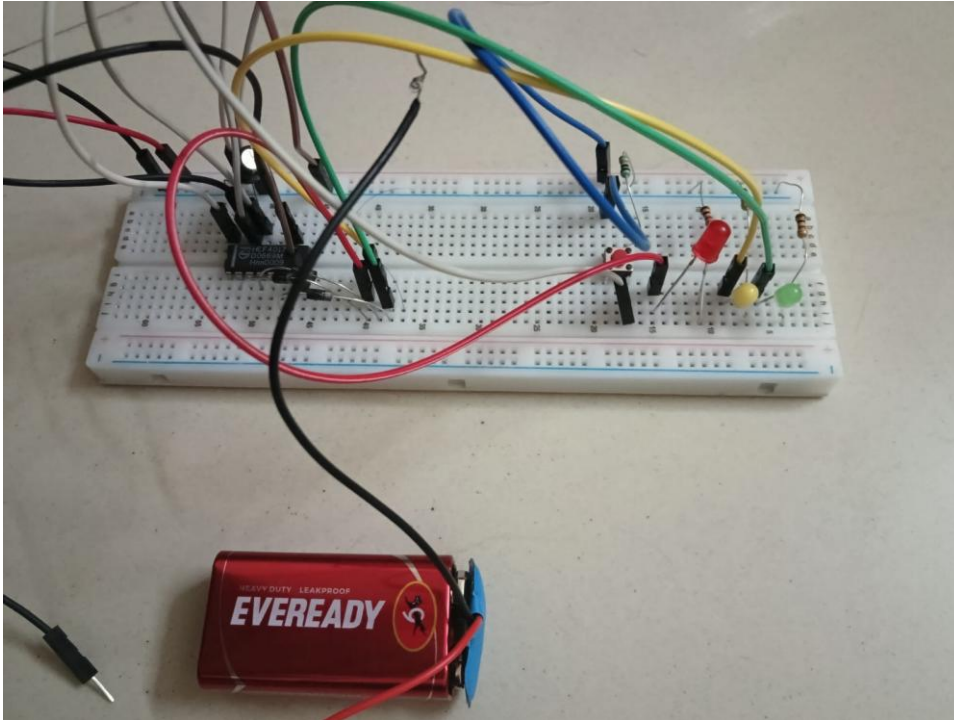
IC Used

74HC4017

Sequence

R → Y → G

Actual Project Hardware



Implementation Highlights

01

Circuit Assembly

CD4017 placed on breadboard; VDD to +9V rail, GND connected.

02

Clock Input

Push button → CLK pin (14). 10k Ω pull-down to GND prevents floating.

03

LED Connection

Q1→220 Ω →Red, Q2→220 Ω →Yellow, Q3→220 Ω →Green → GND.

04

Auto Reset

Q4 (pin) connected to MR (pin 15) for automatic count reset.

05

Testing

Each button press advances signal:
Red→Yellow→Green→Reset cycle confirmed.

✓ Conclusion

Successfully Implemented

A fully functional traffic light controller was built using the CD4017 decade counter IC with Red, Yellow, and Green LEDs cycling automatically.

Objective Achieved

The project achieves hardware-only sequential signal control — no microcontroller or software required.

Digital Design Validated

Truth table, K-map simplification, and Boolean equations correctly describe the circuit behavior confirmed during testing.

Educational Outcome

Hands-on understanding gained in sequential counters, clock pulses, LED driving circuits, and breadboard prototyping.



Future Scope



Microcontroller Upgrade

Replace CD4017 with Arduino/ESP32 for programmable timing and remote monitoring via WiFi.



IoT Integration

Cloud-connected traffic management with real-time data dashboard and emergency override control.



Vehicle Detection

Add IR/ultrasonic sensors to detect vehicle density and adapt green signal timing dynamically.



AI-Based Control

Machine learning models to predict peak traffic hours and optimize signal timing automatically.



Solar Powered

Integrate solar panels with battery backup for eco-friendly, grid-independent operation.

Team Members & Contribution

Soham Ramshette

Roll No: 202501100069

Contribution:

1. Assisted in breadboard setup and component arrangement
2. Helped in wiring connections and checking outputs
3. Supported in testing LED sequence and debugging errors
4. Contributed to project presentation and documentation



Siddhi Dhokale

Roll No: 202501100071

Contribution:

1. Designed and assembled the traffic light controller circuit using CD4017 IC
2. Connected LEDs, push button, resistors, and power supply
3. Helped in circuit testing and troubleshooting
4. Explained the working logic of the project

